

JEE MAIN REVISION MASTER: BIOMOLECULES

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 27 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Amino Acids, Proteins & Nucleic Acids	19 PYQs	Proteins & Genetics
High	Carbohydrates: Monosaccharides, Disaccharides & Glycosidic Linkages	8 PYQs	Carbohydrates

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Amino Acids, Proteins & Nucleic Acids	$\text{Chargaff's Rule: } [A] = [T], [G] = [C], \text{Nucleotide} = \text{Base} + \text{Sugar} + \text{Phosphate}$
Carbohydrates: Monosaccharides, Disaccharides & Glycosidic Linkages	$\text{Sucrose (Non-reducing disaccharide)} = \alpha\text{-glucose} + \beta\text{-fructose} \text{ via } C_1-C_2 \text{ linkage}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Amino Acids, Proteins & Nucleic Acids (Very High Yield — 19 PYQs)

Core Principle: Amino acids exist as dipolar zwitterions. Glycine is the only optically INACTIVE natural amino acid (no chiral carbon). Peptide bond is $-\text{CO}-\text{NH}-$. Denaturation of proteins destroys secondary and tertiary structures (coagulation of egg white), but leaves primary structure intact. DNA contains bases Adenine (A), Guanine (G), Cytosine (C), Thymine (T); RNA contains Uracil (U) instead of Thymine. Chargaff's rule in DNA: $A = T$ (2 hydrogen bonds) and $G \equiv C$ (3 hydrogen bonds).

■ **Caution / JEE Pitfall:** In denaturation of proteins, primary structure (the sequence of amino acids) remains completely UNCHANGED!

■ **Speed Trick:** Water-soluble vitamins are Vitamin B-complex and Vitamin C (cannot be stored in body, excreted in urine); Fat-soluble vitamins are A, D, E, and K (ADEK).

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Incorrect statement about α -amino acids of proteins among the following is..." [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): "Match List-I with List-II List-I (Vitamins) List-II (Deficiency Diseases) (a) B_{12} (i) Convulsions (b) B_{12} (ii) RBC def..." [Ans: N/A]
- [2025 Shift] Q#1 (MCQ): "The correct sequence of α -amino acids, hormone, vitamin, carbohydrates respectively is..." [Ans: N/A]

3.2 Carbohydrates: Monosaccharides, Disaccharides & Glycosidic Linkages (High Yield — 8 PYQs)

Core Principle: Glucose: aldohexose, exists in α and β pyranose cyclic hemiacetal forms. Fructose: ketohexose, forms furanose ring. Disaccharides linked via oxide glycosidic bond: Sucrose (α -D-glucose + β -D-fructose, non-reducing sugar, hydrolysis produces 'invert sugar'); Maltose (two α -D-glucose units, reducing sugar); Lactose (β -D-galactose + β -D-glucose, reducing sugar). Starch is polymer of α -glucose (amylose + amylopectin); Cellulose is linear polymer of β -glucose.

■ **Caution / JEE Pitfall:** Sucrose is dextrorotatory, but its hydrolysis product mixture is LEVOROTATORY because the levorotation of fructose (-92.4°) exceeds the dextrorotation of glucose ($+52.5^\circ$) implies Invert sugar.

■ **Speed Trick:** All monosaccharides (whether aldoses or ketoses) are REDUCING sugars and give positive Tollens' and Fehling's tests.

Representative Source PYQs:

- **[2026 Shift] Q#2 (MCQ):** "Consider the following statements: Statement-I: All monosaccharides are reducing sugars. Statement-II: Sucrose can reduce ammoniacal silver ..." **[Ans: N/A]**
- **[2025 Shift] Q#2 (MCQ):** "Which examples of carbohydrates exhibit α -link, (α -glycosidic link) in their structure?..." **[Ans: N/A]**
- **[2025 Shift] Q#4 (MCQ):** "Match List-I with List-II for the following reaction pattern Glucose $\xrightarrow{\leq}$ Product $\xrightarrow{\text{longright}}$..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Incorrect statement about α -amino acids of proteins among the following is

- (A) Methionine is an essential amino acid
- (B) Glycine doesn't exhibit enantiomerism
- (C) Glycylalanylglycine has three amide linkages
- (D) Zwitterion of valine exhibits amphoteric behaviour

Correct Answer: None

Solution Analysis: The incorrect statement is: Option C Glycylalanylglycine is a tripeptide made from three amino acid units: Glycine-Alanine-Glycine. In a peptide, two amino acids join by one peptide bond, also called an amide linkage. So, for three amino acid units, the number of peptide bonds is: $3-1=2$. Hence, glycylalanylglycine has only two amide linkages, not three. The other statements are correct: Methionine is an essential amino acid. Glycine does not show enantiomerism because its α -carbon has two hydrogen atoms, so it is not chiral. Valine exists as a zwitterion.

Benchmark Problem 2: KCET 2026 (Q#2)

Consider the following statements: Statement-I: All monosaccharides are reducing sugars. Statement-II: Sucrose can reduce ammoniacal silver nitrate solution. Choose the correct answer from the options given below.

- (A) Both Statement I and Statement II are correct
- (B) Both Statement I and Statement II are incorrect
- (C) Statement I is correct but Statement II is incorrect
- (D) Statement I is incorrect but Statement II is correct

Correct Answer: None

Solution Analysis: Statement-I is correct. All monosaccharides are reducing sugars because they have a free aldehydic group or ketonic group, or they can form such a group in solution. So they can reduce reagents like Fehling's solution and Tollens' reagent. Statement-II is incorrect. Sucrose is a non-reducing sugar. In sucrose, the glycosidic linkage is formed between the anomeric carbon of glucose and the anomeric carbon of fructose. So, no free aldehydic or ketonic group is available to reduce ammoniacal silver nitrate solution, also called Tollens' reagent. Hence, the correct answer is: Option C: Statement I